VINEETH RAGHAV GURURAJAN

Date of Birth: 27/05/2004 Nationality: Indian Gender: Male

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PROFILE

M.Sc. Electrical Engineering student at the University of Stuttgart (specialisation Electrical Power Systems) with a B.Tech. in Electronics and Computer Engineering (CGPA 8.70/10). Project experience in battery modeling and power electronics, circuit design and simulation (PSpice), PCB layout (Altium), FPGA design in Verilog, and machine learning with Python. Structured, analytical and detail-oriented; English C1. Seeking a working student role in electrical or hardware development, simulation, power systems or energy data. Available up to 20 hours per week.

EDUCATION

10/2025 – 09/2027 (expected)
M.Sc. Electrical Engineering, University of Stuttgart
Specialisation Electrical Power Systems: Intelligent Power Grids, Power Systems Engineering, Battery Modeling

2021 – 2025
B.Tech. Electronics and Computer Engineering, SRM Institute of Science and Technology, Chennai
CGPA 8.70/10. Grades A+ or A in FPGA-Based Embedded Systems, Embedded Hardware and Operating Systems, Microcontrollers and Interfacing, Hardware Interfacing and Networking, IoT System Design, Deep Learning; final-year project graded A+ (10 credits). Seminar I (2023, grade O): cryptocurrency price analysis and prediction, review of forecasting methods and evaluation metrics (MAE, RMSE).

PROJECTS

2026
Battery Cell Modeling, team project, University of Stuttgart
- Analysed cell-to-cell variance caused by electrical, thermal, aging and manufacturing tolerances by simulation.
- Implemented and debugged statistical parameter sampling in the simulation model.

2025
Optimized 64-bit RoBA Multiplier with MAC Unit, B.Tech. final-year project
- Designed a rounding-based approximate multiplier with a multiply-accumulate unit in Verilog; shift-and-add replaces partial-product computation.
- Synthesised in Xilinx ISE (Spartan-3 target); reported critical path 0.575 ns.

2024
Duplicate Question Detection, team project (4 students)
- Classified duplicate question pairs from the Quora Question Pairs dataset using NLP and fuzzy-ratio features.
- SVM reached 88.3% and Random Forest 80.02% accuracy; deployed as a web application on Heroku.

SKILLS

Electronics and embedded: microcontrollers (ARM, ESP32, Arduino), FPGA design (Verilog, Xilinx ISE), PCB design (Altium), circuit simulation (PSpice), CAN, I2C, SPI, UART

Software and modeling: Python, C/C++, MATLAB, machine learning, statistical parameter sampling

Power and control: power electronics, control systems, sensors and actuators, battery modeling

LANGUAGES

English: C1 (IELTS Academic, overall band 7.5, October 2024) **German:** A2, preparing for the telc B1 exam
Kannada: native

CERTIFICATES AND VOLUNTEERING

PSpice Fundamentals (TVS India); Certificate of Training (Livewire Ltd.). STEM workshops in electronics and coding for underprivileged students; post-cyclone relief work in Chennai.